The Metabolic Module in Three Pages

The flux layer, the parameters it is fed,

and what has been measured

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September 10, 2026

Abstract

The metabolic module is a differentiable layer that maps a perturbed gene representation to a flux vector over yeast-GEM 9.0.2 under stoichiometry, an enzyme-capacity bound and a second-law penalty. Its three parameter tables, turnover numbers, Michaelis constants and formation energies, are meant to be tabulated rather than learned. The tables have been built and audited: six published k_{cat} predictors disagree on a single gene by a median factor of 19, yet 56 to 84 percent of their values land inside the spread of enzymes sharing a function, against 53 percent for a lookup that knows only the function. No trained arm consumes one yet, and the one training grid run so far, 144 cells on Delta, produced constant predictors in every cell, so the module's effect on betaxanthin prediction is unmeasured rather than negative. Full accounts are in the 026 and 027 documents and the flux-layer note.

1 The flux module and what feeds it ✗

The cell graph transformer's instruments map representation to representation (Type I) or to a measured output (Type II). The flux module is neither: it leaves gene index space for reaction space $\mathbb{R}^{4131}$ under a fixed stoichiometry, and its output ν is a latent physical state phenotypes are read from (Figure 1). A deletion sets a gene's availability to zero as a hard fact; the GEM's gene-protein-reaction rules fold gene gates to a reaction by a softmin over a complex's subunits and a sum over isozymes, and the result scales an enzyme-capacity ceiling,

$$|\nu_j| \le c_j(H_{\text{pert}})\, k_{cat,j}\, \bar{E}, \qquad \mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{th}} \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \frac{[\nu_j\, \Delta_r G_j + \epsilon]_+}{|\nu_j|\,|\Delta_r G_j| + \epsilon}. \tag{1}$$

The capacity bound is the only path by which a genotype reaches a flux magnitude, since 4,129 of the GEM's 4,131 bounds are ±1000 or 0; $\bar{E}$ is constant and no enzyme abundance is predicted. The second-law term charges for flux running uphill and, because $\Delta_r G$ sums to zero around any cycle, removes with no binary variable the 427 internal loops a growth constraint leaves free. It acts at training time only; the violation fraction is reported at evaluation. Gibbs dissipation is a logged readout, not a penalty: the Niebel 2019 limit is an empirical plateau, not a law.

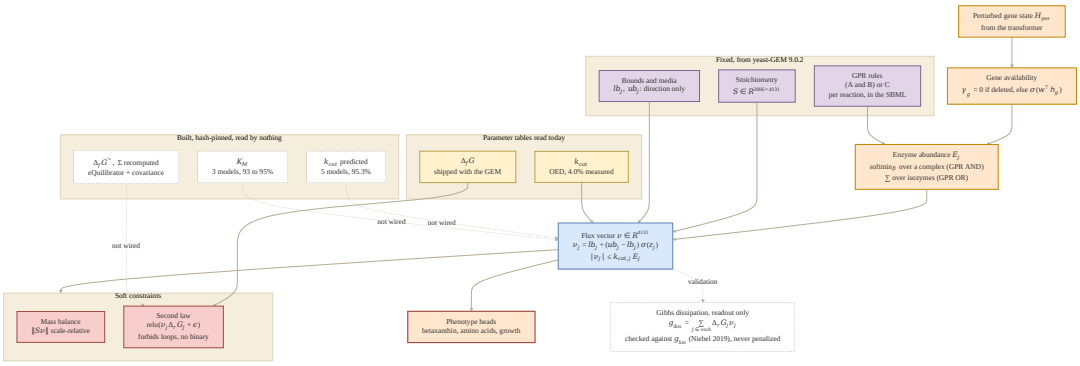


Figure 1: The flux module's main components and what feeds them. Orange, the path from the perturbed gene state to enzyme availability; purple, the fixed yeast-GEM 9.0.2 stoichiometry and bounds; yellow, the parameter tables, a **dashed** border marking one built and read by nothing; blue, the flux vector; red, the soft constraints. Three of the five tables are dashed: k_{cat} resolves from the Open Enzyme Database at 3.7 percent of catalytic units and $\Delta_f G$ from the table shipped with the GEM. A predicted k_{cat} tier and a merged energy route exist in the resolvers, under test. Source `notes/torchcell.models.equivariant_cell_graph_transformer.mermaid.metabolic-module-simple.md`.

2 What the parameter tables are worth ✗

Six published k_{cat} predictors (DLKcat, UniKP, EITLEM, TurNuP, DeepEnzyme, EZKinetics) and four K_M predictors were run from hash-pinned mirrors over 95.3 percent of yeast-GEM's 3,728 catalytic units, the ceiling set by substrate structure availability. Per gene, the unit the capacity bound reads (Figure 2), the median gene moves 1.28 orders of magnitude between the highest and lowest predictor and 71 percent move by more than one order, against a consensus range across genes of 0.87. On the 57 measured yeast pairs, EZKinetics leads on rank at Spearman 0.79 and EITLEM on magnitude at 0.23 orders median error, an upper bound: every predictor trained on the literature those measurements come from.

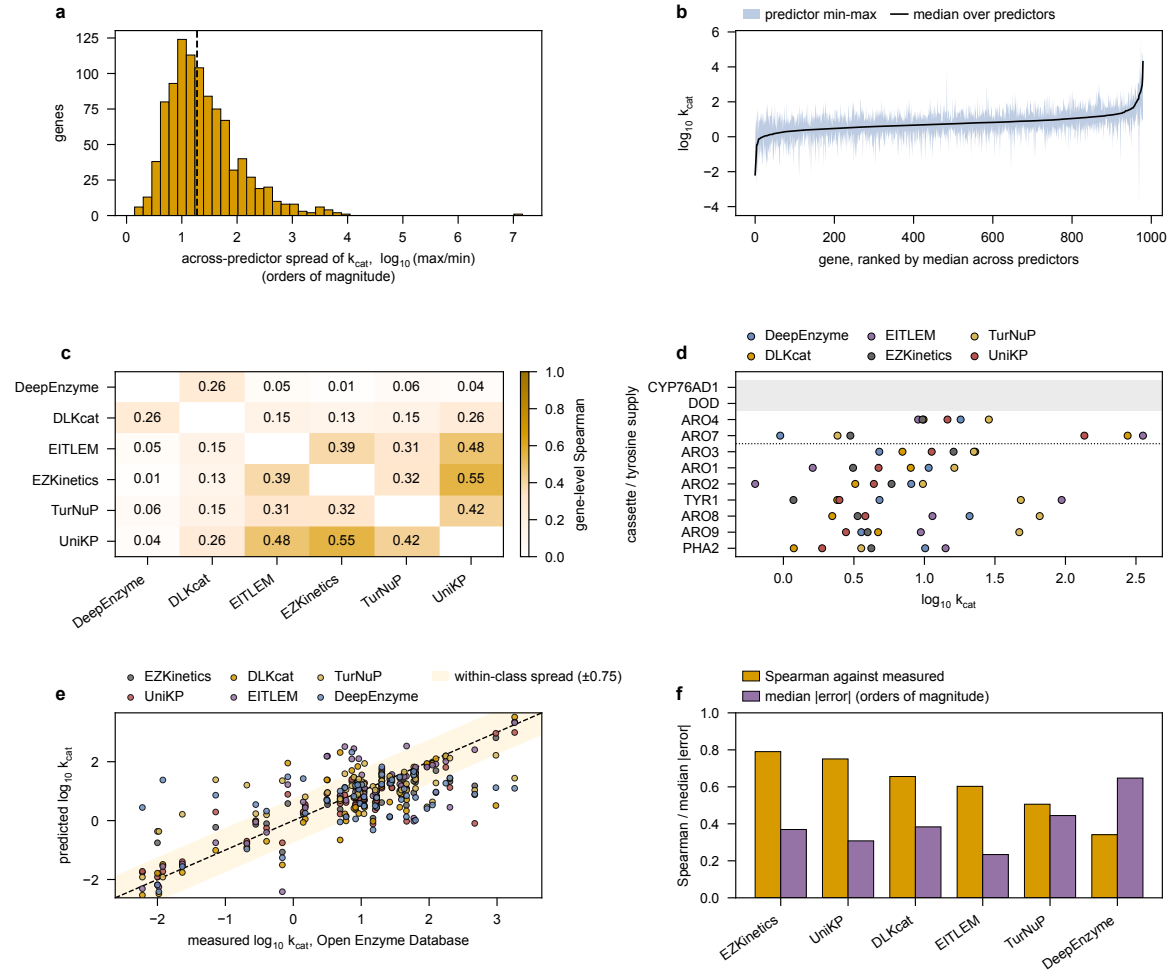


Figure 2: Agreement between the six k_{cat} predictors, and their accuracy against measured yeast values. Panels a to d cover the 980 genes all six predictors cover; e and f the 57 measured pairs. **a**, Across-predictor spread per gene; dashed, the median of 1.28 orders of magnitude. **b**, The same spread against the dynamic range of k_{cat}, genes ranked by their median. **c**, Gene-level Spearman between predictors, 0.01 to 0.55. **d**, The betaxanthin supply genes; the shaded band marks the two heterologous cassette members with no yeast-GEM reaction. **e**, Predicted against measured; the band's half width of 0.75 is the median standard deviation of $\log_{10} k_{cat}$ within one EC4 class. **f**, Spearman and median absolute error, ranked. Figure 6 of the 026 document, from `plot_per_gene_predictor_agreement.py`.

Most of that plausibility is coarse-class knowledge (Figure 3). Against the EC4 band every predictor beats the EC-median lookup, 56 to 84 percent against 53; at EC1 the lookup already reaches 79 percent, and for K_M no predictor beats the lookup at all. On the nine unmeasured betaxanthin supply genes, four of six k_{cat} predictors are class-plausible and no K_M predictor is.

The thermodynamic table is recomputed with eQuilibrator component contribution, which returns a covariance the shipped point values cannot. It prices 2,741 of 4,131 reactions, 986 recovered by fixing the proton convention; with the shipped table it reaches 3,716, 94.4 percent of flux-carrying reactions. **The shipped file is in kilocalories and labeled kilojoules**, copied verbatim from yETFL by yeast-GEM's curation script: regressing our table on yETFL's over 1,927 shared metabolites gives a slope of 4.187 at $r = 0.987$. Correcting that and matching compartment conditions closes the transport arm of the disagreement, median 17.4 to 5.6 kJ mol^{-1}, leaving the single-compartment cloud at 17.6, the genuine estimator difference. Of the 665 unpriced metabolites, 185 carry placeholder formulas and are a method ceiling;

347 of the rest now carry a structure written from the name and a price, taking the recomputation past the shipped column to 86.3 percent, though the union does not move and the values need re-transforming per compartment. All five betaxanthin intermediates resolve by InChIKey, and Uni-pKa on its simple enumeration template reproduces ChemAxon's energies to 0.91 kJ mol^{-1}, so the license binds nowhere.

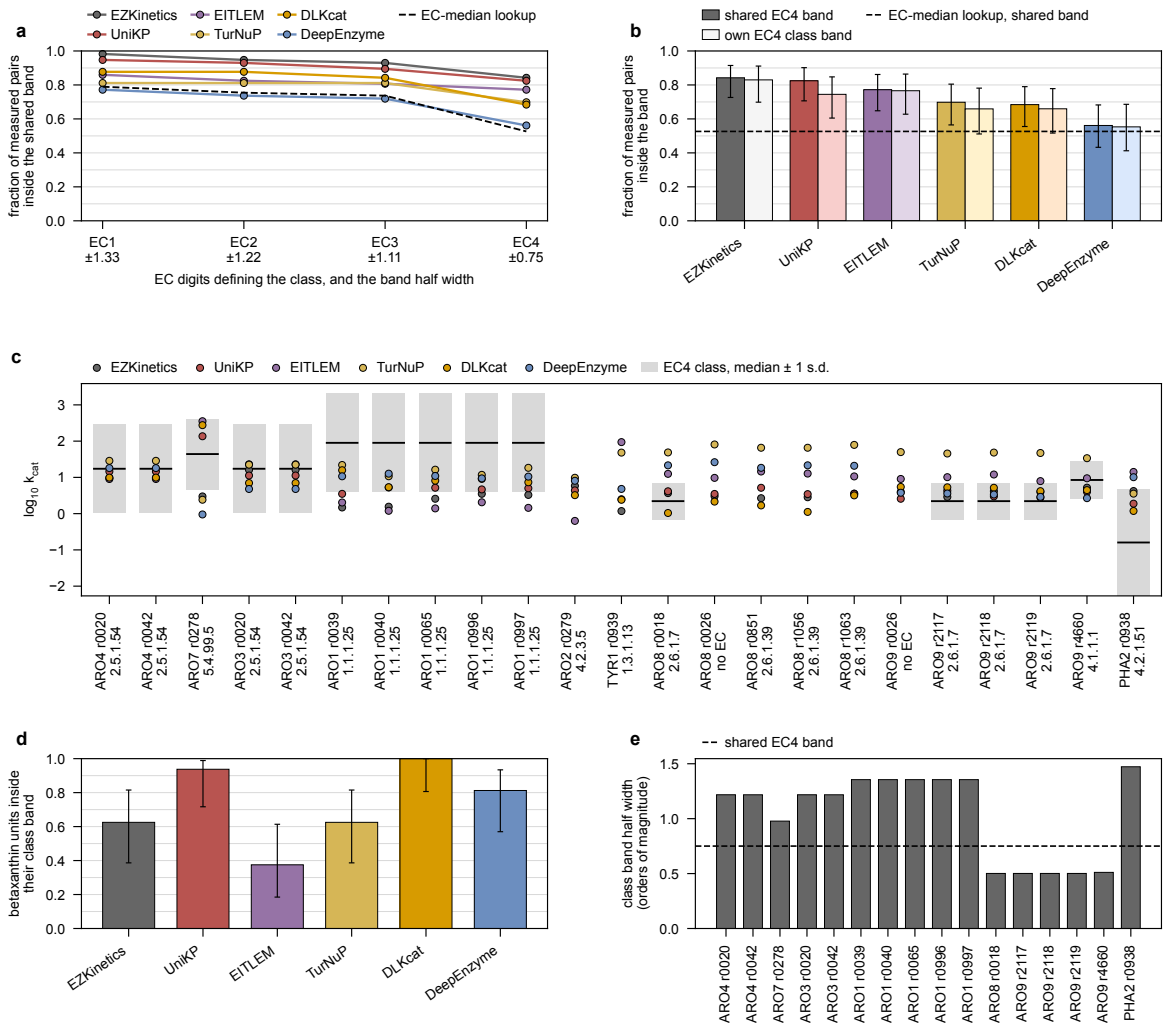


Figure 3: How many of each predictor's k_{cat} values fall inside the spread of enzymes doing the same job. a, Fraction of the 57 measured pairs inside a shared band as the class is read to fewer EC digits; dashed, the EC-median lookup. **b**, The EC4 fraction against each pair's own class band, Wilson 95 percent intervals. **c**, The 23 betaxanthin supply units against one standard deviation of their EC4 class. **d**, Fraction of the 16 scorable units inside, by predictor: DLKcat 16 of 16, UniKP 15, EITLEM 6. **e**, Each unit's class band half width. Figure 7 of the 026 document, from `plot_within_class_plausibility.py`.

3 Where the training stands ✗

The 027 grid on Delta ran 144 cells, six arms over 24 seeds, on Cachera's betaxanthin screen with the 019 nine-graph backbone. The anchored module scored −0.020 test Spearman against +0.027 for no module. The prediction dumps show every cell is a constant predictor: the largest prediction spread in any cell is 0.8 percent of the target spread, the no-module arms sit at the target mean with a training loss of exactly 1.00 from epoch 10, and the flux arms at wrong constants. The claim that the thermodynamic anchor costs is therefore withdrawn and the module's effect is unmeasured; the same backbone reached 0.434 under the 019 trainer, so the defect is in the 027 loop. Two defects precede any relaunch: training uses yeast-GEM's shipped exchange bounds, which forbid all amino-acid uptake, while the Cachera strains grew in a supplemented medium; and 170 of the 183 proton-translocating reactions carry a confidently wrong zero, the translocation term being computed but off by default.

Order of work. A trainer that reproduces 0.434 on the no-module arm with an abort on prediction collapse; the module arms on it, on the dataset's actual medium; then the merged $\Delta_f G$ and predicted k_{cat} tables, on GilaHyper jobs 1663, 1667, 1668 and 1669, carried to a selected epoch.